UTF-8 Support Notes

Bruce Rennie

Abstract

Application notes on Unicon UTF-8 string support, operator overloading, and related class libraries.

Unicon Project
http://unicon.org
University of Idaho
Department of Computer Science
Moscow, ID, 83844, USA

1 UTF-8 Support Notes

Application notes on Unicon UTF-8 string support, operator overloading, and related class libraries.

Notes by Bruce Rennie.

In relation to mapping lower to upper case or upper to lower case, there are a number of difficulties that will be encountered. These are documented on the Unicode.org website. To that end, three files have been copied from Unicode.org and placed under `assets/utf8/` for reference in developing appropriate mapping regimes. These mapping sets will need to be on a language by language basis and there are situations where there is no 1-1 mapping available for lower to upper or upper to lower case.

There is now the corresponding methods and operator overload facilities for dealing with the use of UTF8 strings as numeric values and allowing these values to be converted to the appropriate numeric values for use in numeric operations.

There are now consistent runtime errors in essentially all places where Unicon has runtime errors occurring for Unicon strings. The error messages are of the same standard format as the underlying Unicon system and use the same standard error codes as found in the Unicon runtime.

1.1 Reference data

Unicode Consortium tables used when developing case-mapping regimes:

- UnicodeData.txt_ — Unicode character database
- CaseFolding.txt_ — case-folding mappings
- SpecialCasing.txt_ — special-casing rules

Case mappings are language-dependent; some letters have no one-to-one upper/lower mapping.

1.2 Package and class inventory

1.2.1 Package `ErrorSystem`

`class ErrorSystem`

```
method __proc_components(level, pos)
method AbortIfDebug(param[])
method AbortMessage(errorcode, file, lineno, param[])
method Caller(addedlvl)
method CallingComponents(name)
method ClassCalling(level)
method Debug(mode)
method DebugMode()
method ErrorOut(errorfile, writeappend)
method IsOn()
method MethodCalling(level)
method NotInDebugMode()
method Off()
method On()
method PackageCalling(level)
method ParamCalling(level)
method PrintMessage(param[])
method ProcedureCalling(level)
method StopMessage(param[])
```

Package-level procedures

```
procedure AllOff()
procedure AllOn()
procedure trace(message[])
```

1.2.2 Package `ClassObject`

`class ClassClass : ErrorSystem`

```
method __InitClass()
method SetClass(obj)
method __AuxMessage(errorno, messageno)

method Addition(val1, val2)
```

```
method Divide(val1, val2)
method Minus(val1, val2)
method Modulus(val1, val2)
method Multiply(val1, val2)
method Negate(val)
method UnaryPlus(val)
method Power(val1, val2)

method Equals(val1, val2)
method GTorEqual(val1, val2)
method Greater(val1, val2)
method LTorEqual(val1, val2)
method Less(val1, val2)
method Nequal(val1, val2)

method Integer(val)
method Numeric(val)
method Real(val)
```

class ClassObject : ErrorSystem

```
abstract method String()
method __call_by_method_name(methname, params[])
method ClassName()
method Clone()
method Copy()
method operator_overload_error_error(methname)
method Serial()
method set_operator_overload_abort(noabort)

method Addition(val)                    -->     __add__
method Divide(val)                      -->     __div__
method Minus(val)                       -->     __minus__
method Modulus(val)                     -->     __mod__
method Multiply(val)                    -->     __mult__
method Negate()                         -->     __neg__
method UnaryPlus()                      -->     __number__
method Power(val)                       -->     __powr__
```

```
method Equals(val)                       -->     __numeq__
method GTorEqual(val)                    -->     __numge__
method Greater(val)                      -->     __numgt__
method LTorEqual(val)                    -->     __numle__
method Less(val)                         -->     __numlt__
method Nequal(val)                       -->     __numne__

method Integer()
method Numeric()
method Real()

method __add__(y)                        -->     x + y
method __div__(y)                        -->     x / y
method __minus__(y)                      -->     x - y
method __mod__(y)                        -->     x % y
method __mult__(y)                       -->     x * y
method __neg__()                         -->     - x
method __number__()                      -->     + x
method __powr__(y)                       -->     x ^ y

method __eqv__(y)                        -->     x === y
method __neqv__(y)                       -->     x ~=== y

method __lexeq__(y)                      -->     x == y
method __lexge__(y)                      -->     x >>= y
method __lexgt__(y)                      -->     x >> y
method __lexle__(y)                      -->     x <<= y
method __lexlt__(y)                      -->     x << y
method __lexne__(y)                      -->     x ~== y

method __numeq__(y)                      -->     x = y
method __numge__(y)                      -->     x >= y
method __numgt__(y)                      -->     x > y
method __numle__(y)                      -->     x <= y
method __numlt__(y)                      -->     x < y
method __numne__(y)                      -->     x ~= y

method __cat__(y)                        -->     x || y
method __lcat__(y)                       -->     x ||| y
```

```
method __compl__()                        -->     ~ x
method __diff__(y)                        -->     x -- y
method __inter__(y)                       -->     x ** y
method __union__(y)                       -->     x ++ y

method __bang__()                         -->     ! x
method __random__()                       -->     ? x
method __refresh__()                      -->     ^ x
method __sect__(y, z)                     -->     x[i:j]
method __size__()                         -->     * x
method __subsc__(y)                       -->     x[i]
method __tabmat__()                       -->     = x
method __toby__(y, z)                     -->     x to y by z
```

Package-level procedures

```
procedure __NewNotSet(val)
procedure AlreadyRun(object_tested)
procedure DeepCopy(A, cache)
procedure Execute(object_or_class_value)
procedure IsObjectOf(value, expectedtypes[])
procedure IsObject(value_tested)
procedure ObjectProperties(object_or_classname)
procedure set_operator_overload_error_messages()
procedure SuperClass(this_object)
procedure HasMethod(val, methodname)
```

1.2.3 Package UTF8

class UTF8 : ClassClass

```
method __Initialise()
method __ActualPos(ustr, i)
method __Coerce(val)
method __GetAt(ustr, i, j)
method __AuxMessage(errorno, messageno)
```

```
method BOMFound(str)
method SkipBOM(str)
method Valid(str)
method ByteTranslate(byte)
method Multibyte(codepoint)
method SpecialString(str)
method ToUnicode32(ustr)
method Unicode32To(unicode)
method HexTo(hexcode)
method DecimalTo(num)

method AmpPos()

method Equiv(val1, val2)
method Nequiv(val1, val2)

method LexEquals(str1, str2)
method LexGTorEq(str1, str2)
method LexGT(str1, str2)
method LexLTorEq(str1, str2)
method LexLT(str1, str2)
method LexNE(str1, str2)

method Concatenate(ustr1, ustr2)

method Complement(val)
method Difference(val1, val2)
method Intersect(val1, val2)
method Union(val1, val2)

method ForEach(str)
method Size(str)
method Random(str)
method Subsection(ustr, i, j)
method Subscript(str, i)
method TabMatch(str)

method Any(uset, ustr, i, j)
method Bal(uset, usetopen, usetclose, ustr, i, j)
method Center(ustr, i, pstr)
```

```
method Char(i)
method Cset(val)
method Detab(ustr, i[])
method Entab(ustr, i[])
method Find(ustr1, ustr, i, j)
method Left(ustr, i, pstr)
method Many(uset, ustr, i, j)
method Map(ustr, ustr1, ustr2)
method Match(ustr1, ustr, i, j)
method Move(i)
method Ord(ustr)
method Pos(i)
method Repl(ustr, i)
method Reverse(ustr)
method Right(ustr, i, pstr)
method String(val)
method Tab(i)
method Trim(ustr, uset, i, allutf8blanks)
method Upto(uset, ustr, i, j)
```

class __UTF8Object : ClassObject

```

method Data()

method Complement()                    -->     __compl__
method Difference(val)                 -->     __diff__
method Intersect(val)                  -->     __inter__
method Union(val)                      -->     __union__

method Equiv(val)                      -->     __eqv__
method Nequiv(val)                     -->     __neqv__

method LexEquals(str)                  -->     __lexeq__
method LexGTorEq(str)                  -->     __lexge__
method LexGT(str)                      -->     __lexgt__
method LexLTorEq(str)                  -->     __lexle__
method LexLT(str)                      -->     __lexlt__
method LexNE(str)                      -->     __lexne__
```

```
method ForEach()                        -->     __bang__
method Random()                         -->     __random__
method TabMatch()                       -->     __tabmat__
method Size()                           -->     __size__
method Concatenate(ustr)                -->     __cat__
method Subsection(i, j)                 -->     __sect__
method Subscript(i)                     -->     __subsc__

method String()
method Invalid()
```

class UTF8Set : ClassClass

```
method __Coerce(val)
method __AuxMessage(errorno, messageno)

method Complement(uset)
method Difference(uset1, uset2)
method Intersect(uset1, uset2)
method Union(uset1, uset2)

method ForEach(val)
method Size(val)
method String(val)
method Subscript(val, i)
method Member(uset, val[])

method Space()
method UTF8Spaces()
method Ascii()
method Rparens()
method Lparens()
method All()
method MaxSize()
method Defined(str)
```

class __UTF8SetObject : ClassObject

```
method __DebugPrint(modifier, val, uset)

method Equiv(uset)                          -->     __eqv__
method Nequiv(uset)                         -->     __neqv__

method LexEquals(str)                       -->     __lexeq__
method LexGTorEq(str)                       -->     __lexge__
method LexGT(str)                           -->     __lexgt__
method LexLTorEq(str)                       -->     __lexle__
method LexLT(str)                           -->     __lexlt__
method LexNE(str)                           -->     __lexne__

method Complement()                         -->     __compl__
method Difference(val)                      -->     __diff__
method Intersect(val)                       -->     __inter__
method Union(val)                           -->     __union__

method ForEach()                            -->     __bang__
method Size()                               -->     __size__
method Subscript(i)                         -->     __subsc__
method Random()                             -->     __random__
method TabMatch()                           -->     __tabmat__
method Concatenate(ustr)                    -->     __cat__
method Subsection(i, j)                     -->     __sect__

method Member(c[])

method UTF8Value()
method String()
```